

PARTICULAR SPECIFICATION P13

Adoption of the Board's Asset Management System

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P13.1 ASSET MANAGEMENT SYSTEM

P13.1.1 Overview

- (a) The Board is implementing a Board-wide Asset Management System (System) to manage the maintenance of its assets. The Contractor shall adopt the System while implementing Operations and Maintenance (OM) activities described in Particular Specification P12 Operations and Maintenance.
- (b) In adopting the System, the Contractor shall comply with all procedures and workflow processes required for using the System as prescribed by the Board. These will be confirmed to the Contractor after award of Contract.
- (c) In adopting the System, the Contractor shall be responsible for the provision of qualified staff, competent in the use of the System and adhering to the procedures and workflow processes. The Contractor shall ensure that his staff deployed are trained and equipped with a basic level of IT competency which includes Microsoft Office and use of the mobile device. For Contractor's staff deemed incompetent by the Board following repeated failure to correctly use the system, the Contractor shall provide a competent replacement staff.
- (d) The Board will provide one (1) round of training to the Contractor's staff who will be using the System. This one-time training will be for up to 20 people.
- (e) In the event of staff changes during the Contract Period, the Contractor shall ensure that all new staff have attended the training program on using the System. It is the Contractor's responsibility to ensure that the new staffs who will be using the System are adequately trained and well-versed in it.
- (f) The Board may arrange for additional rounds of training upon request by the Contractor. However, the Board reserves the right to reject the request if it is deemed unreasonable (e.g. too frequent, repeated failure to attend). Any such rejection shall be without prejudice to the Contractor's responsibility to comply with the requirements of the Contract.
- (g) The Contractor will provide the asset information in the AMS using the template provided by the Board during the Contract Period.
- (h) The typical work process to be followed by the Contractor in using the System includes:
 - 1) Generate job cards (to be agreed with the Board) wirelessly/remotely on their device;
 - 2) Assign job card to the Contractor's staff;

- 3) Provide Status updates (e.g. “Acknowledge”, “Received Job”, “Reached Site”, “Completed Job”) of job cards from job location, including scanning of RFID tags; and
 - 4) Make submissions (e.g. checklists, photos/videos, service reports, update quantities in FSOR) for each job card.
- (i) The Board may modify the above workflow to allow the Contractor to receive and provide updates on a periodic basis (e.g. daily) instead of in real-time.
 - (j) The Board reserves the right to make changes to the System, including the workflow processes to be followed by the Contractor, at any time. The Contractor shall readily adopt and fully comply with such changes and at no additional cost to the Board.
 - (k) The Contractor shall use the System to generate softcopies of the deliverables (e.g. field inspection checklists and field reports) that have been submitted directly to the System. The Contractor shall generate hardcopies of these deliverables, review it, attest it, and submit to the Board. A Works Order shall be deemed to have been complete only upon fulfilment of this requirement. For example, field or inspection checklists will be submitted by the Contractor from the field using the System. The Contractor shall generate softcopies of these site checklists using the System, review these site checklists for factual accuracy, attest it, and submit it to the Board together with the other deliverables for that Works Order.
 - (l) Notwithstanding submission of checklists/reports/updates/notifications using the system, the Contractor shall follow the requirements to notify the Board (e.g. scenarios and issues that require urgent attention) as per the procedures and service levels in the Specifications.
 - (m) The Contractor shall note that the Board reserves the right to reject any job, checklist, reports, and photo/videos (in part or whole) that the Board may reasonably find to be not acceptable. It shall be the Contractor’s responsibility to make good all rejected work at no additional cost to the Board.
 - (n) The System has a functionality to send alerts / reminders by SMS. The Contractor shall provide the staff’s mobile phone number to be configured in the System for the receipt of relevant messages.
 - (o) The Contractor’s staff shall proactively address all electronic alerts (e.g. preventive maintenance activity alarm criteria exceeded) issued by the System to their mobile phone. The Contractor shall proactively seek the Board’s approval to proceed with the Contractor’s recommended rectification.

- (p) After the adoption of the System, if the Contractor is unable to access the System due to system downtime or for whatever reason, the Contractor shall revert to manual work procedures (using hardcopy forms and Excel). The Contractor's staff shall know the manual maintenance work procedures. The Contractor shall be prepared to adopt this as a contingency plan. The Contractor shall always maintain and provide the necessary resources to implement the contingency plan immediately. The Contractor shall also update the data for the works carried out into the System – directly or using excel templates, as required by the Board. The Contractor shall comply with this requirement at no additional cost to the Board.

P13.1.2 Hardware Requirements

- (a) The Contractor shall use his own mobile devices for the purpose of accessing and using the System.
- (b) The Contractor's staff shall be issued a mobile device with data plan to be supplied by the Contractor as per the minimum requirements in P13.1.7.
- (c) The Contractor shall not commence any procurement related to adoption of the System unless prior written approval from the Board has been obtained.
- (d) In adopting the System, the Contractor shall provide the following items for use by their staff (including field teams) upon instruction from the Board:
 - 1) Required number of mobile devices (as described below) - each of the Contractor's field team shall be equipped with a mobile device. The mobile devices may be shared provided it meets the requirements of the System and does not compromise the efficiency of the works carried out;
 - 2) Accessories that may be required for the mobile device (spare battery, carry sleeve, protective casing, and charger);
 - 3) Necessary maintenance, repair, and support for the mobile device, including stand-by units; and
 - 4) Subscribe to a suitable commercially available wireless data plan for the mobile devices (minimum of 3GB per month).
- (e) The Contractor shall own and be responsible for the security and maintenance of the items and ensure that they are in proper working condition during the Contract Period. Any damage to the items during the Contract Period that warrant a repair or service cost shall be borne by the Contractor.
- (f) The Contractor shall ensure that repairs/maintenance of the items are done promptly without impact on staff's use / adoption of the System.

- (g) The Contractor shall allow the Board to affix Velcro or other attaching mechanism to the mobile device (or its protective casing). This is for the purpose of attaching the mobile device to the RFID reader.

P13.1.3 No Separate Payment for Adoption of AMS

- (a) There shall be no separate rate / payment for adoption of the System including the use of the items in P13.1.2. Any costs incurred by the Contractor in relation to the adoption of the System is deemed inclusive in the rates for carrying out the scope of works in this Contract.

P13.1.4 AMS Software on the Mobile Device

- (a) The Contractor's mobile device will be setup with a web-based and/or client application that will provide access to the System. The devices used by the Contractor's field teams shall be connected to the System in real-time using the installed application, via a commercially available wireless data plan. The Contractor shall ensure that the relevant Contractor's staffs are logged in to the System (via the device) continuously during their operating hours under this contract, as required by PUB.
- (b) All software licenses, updates, and upgrades to the AMS software application installed on the devices will be provided by the Board at no charge to the Contractor.
- (c) The Board may also install other software to ensure / enforce security requirements and manage the AMS software application and related data on the mobile device.

P13.1.5 IT Requirements

- (a) The Contractor and its staff shall comply with all IT standards and guidelines, including security requirements, issued by the Board. The Contractor shall not compromise the Board's data, network and information systems.
- (b) At the end of the Contract Period, the Contractor shall promptly ensure that all data, information, photographs, documents, databases, etc. pertaining to/generated during this contract are deleted and/or returned to the Board, as required by the Board.
- (c) At the end of the Contract, all software installed by the Board and related data on the device will be uninstalled and deleted by the Board. The Contractor shall facilitate this process at no additional cost.

P13.1.6 Additional Requirements for any Hardware Issued by the Board

- (a) During the Contract Period, the workflow process requires the scanning of RFID tags, the Contractor shall use the RFID reader(s) under the provision of Contract to attach to the mobile device.

- (b) The Board may restrict the availability of these items to the Board's premises only. For items with such restriction, the Contractor staff shall borrow the item at the Board's premises, use it at the Board's premises, and then return it before leaving the Board's premises. The Contractor and its staff shall comply with the procedures imposed by the Board in relation to the access and use of these items.
- (c) The Contractor shall comply with the Board's requirements for the use of such items at no cost to the Board.
- (d) The Contractor shall make the device available to the Board (or its Contractors) to check and maintain the items as and when required by the Board at one of the Board's premises at no cost to the Board.
- (e) The Contractor shall undertake all effort to safeguard all such items that may be issued to the Contractor by the Board.
- (f) In the event of any loss, tampering or damage to such item, the Contractor shall immediately report it to the Board. The Contractor shall bear the cost of repair or replacement of the item. The Contractor shall comply with the necessary procedures prescribed by the Board for reporting of such incidents and cost recovery. The Board reserves the right to impose administrative charges for costs and expenses incurred by the Board due to the event and will be deducted from any monies that are due or become due to the Contractor.
- (g) At the end of the Contract, the Contractor shall handover two (2) new RFID reader(s) to the Board.

P13.1.7 Minimum Requirements for Mobile Devices

- (a) The **minimum requirements** for the mobile device to be provided and used by the Contractor to adopt the Board's AMS are as follows:
 - 1) 9.7 – 12.9 inch (diagonal) LED-backlit Multi-touch display with IPS technology;
 - 2) 2160 x 1620-pixel resolution;
 - 3) 32 GB storage space;
 - 4) Fingerprint resistant oleophobic coated and antireflective screen;
 - 5) A12 chip with 64-bit architecture;
 - 6) M8 motion coprocessor;
 - 7) 8 Megapixel camera;

- 8) Wi-Fi (802.11a/b/g/n/ac); dual band (2.4GHz and 5GHz); HT80 with MIMO;
- 9) Operating System iOS 9;
- 10) Built-in 27.3-Whr rechargeable lithium-polymer battery;
- 11) Up to 9 hours of surfing the web using a mobile data network; and
- 12) Accessories for each mobile device may include: spare battery with 6 hours battery lifespan, carry sleeve, designed for mobile workflow with compatible casing, adjustable shoulder strap for hands-free, docking station.

END OF SECTION